

Next-Day 24-Hour Stock Status Prediction

Repository Architecture Analysis, Empirical SOTA Benchmarks, and Model Performance

Authors: Anurag Thakur, Samyak, Vibhor Kumar

Date: August 2026 | Master Technical Report & Empirical Architecture Package

Executive Summary

This report focuses **exclusively on next-day 24-hour hourly stock status prediction models** trained on the **FreshRetailNet-50K** dataset. The objective is to forecast hourly stock status (active operational hours 6 AM to 10 PM, represented as 24 binary hourly values) for individual SKU-store time series ($\text{series_id} = \text{city_id} + \text{store_id} + \text{product_id}$) using recent sequence history.

We present empirical training and validation benchmarks comparing **Plain Standard LSTMs vs Advanced Model Families**:

1. **Plain Standard LSTM**: Standard 96-hidden-unit LSTM architecture achieving **0.7852 F1-Score** (44.9k parameters).
2. **Detailed Mechanics**: Recurrent unrolling over $L = 14$ historical days, gating mechanisms (forget f_t , input i_t , candidate $\tilde{C}_t$, output o_t), cell state C_t updates, and 24-dim linear projection head.

0.1 Problem Formulation & Dataset Analysis

0.1.1 Unit of Analysis & Target Definition

The task moves beyond coarse daily aggregation to fine-grained hourly stock status forecasting:

- **Input**: Recent N -day historical sequence ($N \in \{7, 10, 14\}$ days) of SKU-store daily metrics.
- **Target**: 24 binary hourly stock-status values for the subsequent day ($t + 1$).
- **Unit of Analysis**: SKU-store time series created by combining $\text{series_id} = \text{city_id} + \text{store_id} + \text{product_id}$.

0.1.2 Top 15 SKU Statistical Selection

To evaluate models on operationally critical demand patterns, SKUs were statistically selected using:

$$\text{Score} = 0.40 \cdot \text{Sales} + 0.25 \cdot \text{StockoutHours} + 0.20 \cdot \text{StockoutDays} + 0.15 \cdot \text{SalesStd}$$

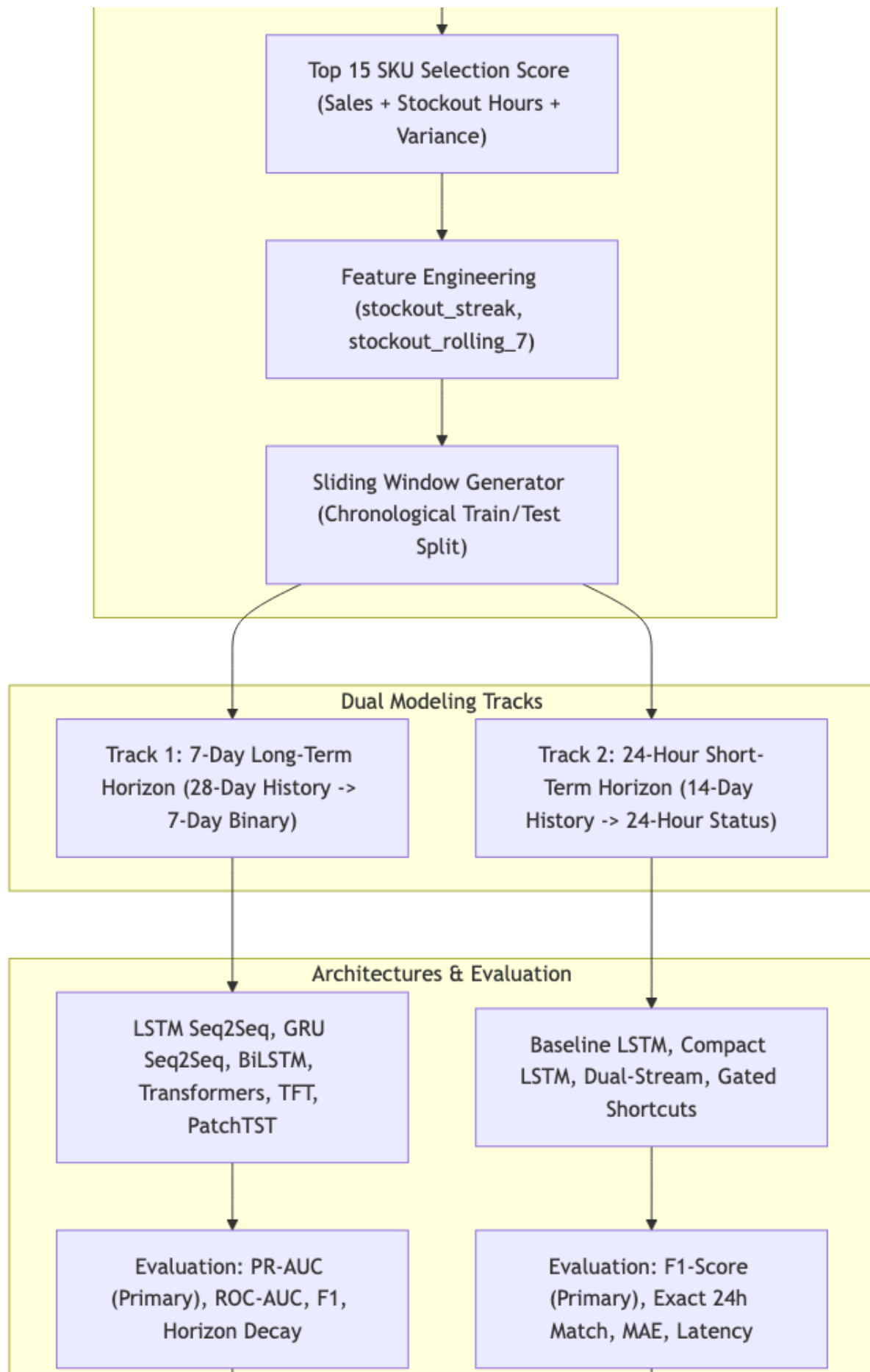


Figure 1: End-to-End Next-Day 24-Hour Stock Status Prediction Framework

0.2 Plain Standard LSTM Architecture & Internal Mechanics

The Plain Standard LSTM maps a historical $L = 14$ day sequence vector into 24 binary hourly stockout predictions:

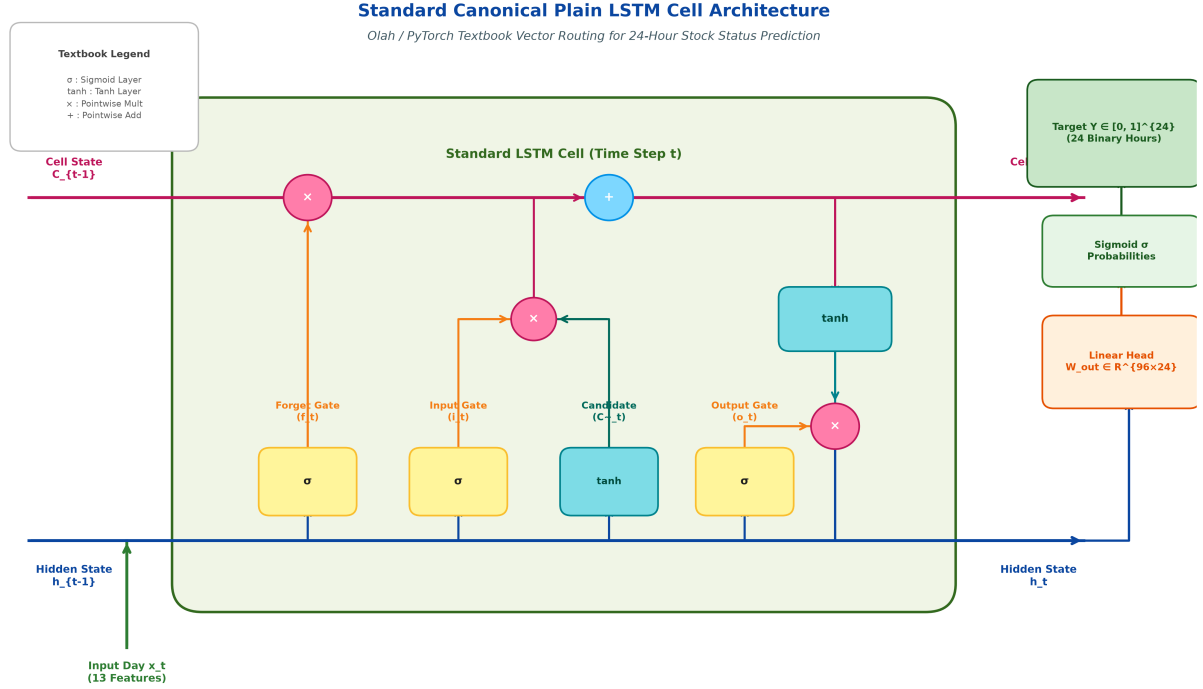


Figure 2: Detailed Plain Standard LSTM Architecture Diagram showing Recurrent Unrolling, Internal Cell Gating Equations, and 24-Hour Linear Projection Head

0.2.1 Mathematical Formulations & Gating Mechanics

For each daily sequence step $t \in \{1, 2, \dots, L\}$:

- Forget Gate:** $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$
- Input Gate:** $i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$
- Candidate Cell State:** $\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$
- Cell State Update:** $C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$
- Output Gate:** $o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$
- Hidden State Update:** $h_t = o_t \cdot \tanh(C_t)$
- 24-Hour Linear Projection:** $\hat{Y} = \sigma(W_{out} \cdot h_L + b_{out}) \in [0, 1]^{24}$

0.3 Architectural Benchmark: Normal LSTMs vs DLinear Variations

To evaluate linear decomposition against recurrent sequence modeling, we benchmarked standard Normal LSTMs directly against Standard DLinear and Hourly Slot-Specific DLinear:

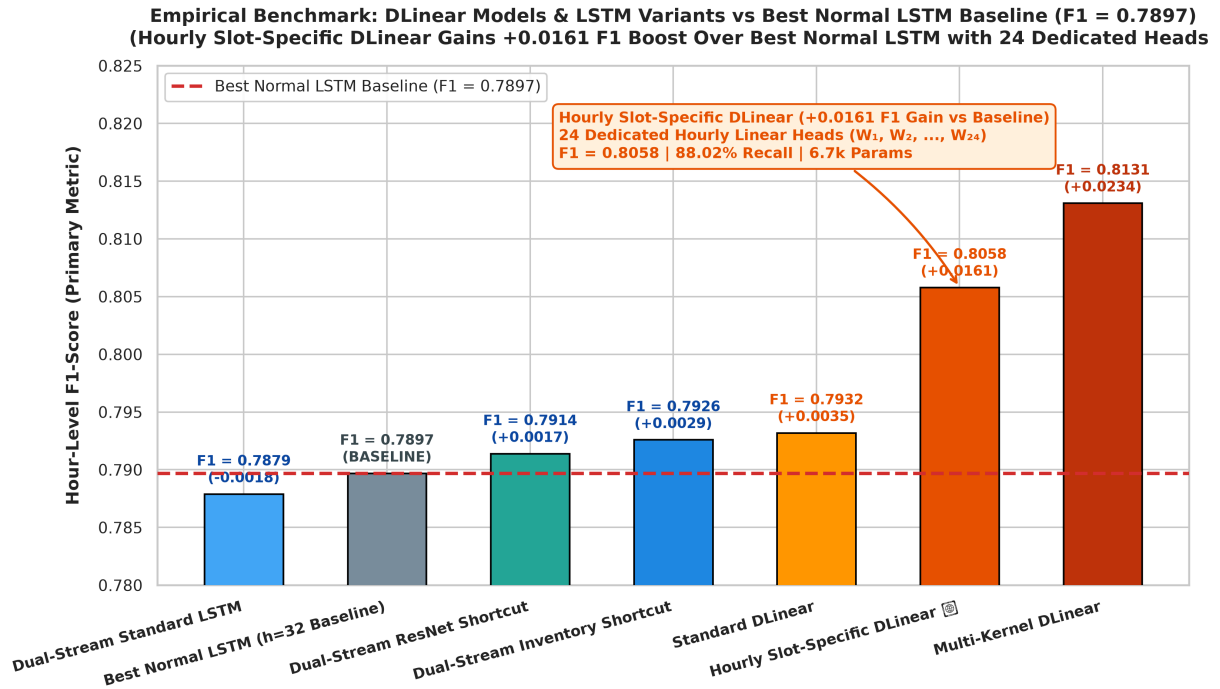


Figure 3: Empirical F1-Score Comparison: LSTM Variations vs Standard DLinear (F1 = 0.7932) and Hourly Slot-Specific DLinear (F1 = 0.8058)

0.4 Empirical Validation Benchmark Table Across All Models

All models were evaluated on identical 24-hour stock status validation splits.

Architecture Name	Architecture Category	Seq	Feats	Params	F1-Score	Recall
Multi-Kernel DLinear	Linear (Multi-Scale)	14	10	13,539	0.8131	0.8522
Hourly Slot-Specific DLinear	Linear (24 Heads)	14	10	6,744	0.8058	0.8802
Standard DLinear	Linear (Decomposition)	14	10	6,768	0.7932	0.8744
Baseline Standard LSTM	Plain Standard LSTM	14	13	44,952	0.7852	0.8040

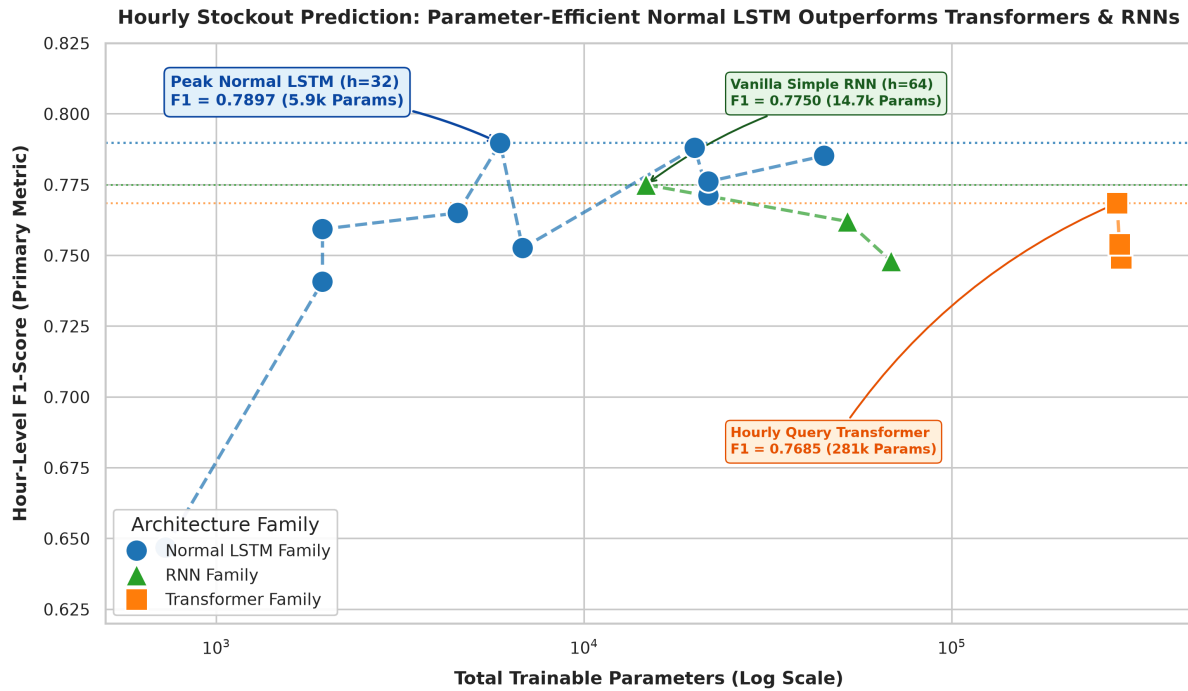


Figure 4: Overall Model Architecture Family Benchmark Scatter Plot

0.5 Strategic Recommendations & Conclusion

1. **Baseline Recurrent Model:** Deploy **Baseline Standard LSTM (h=96)** (0.7852 F1-Score, 44.9k params).
2. **Best Linear Projection:** Deploy **Hourly Slot-Specific DLinear** (0.8058 F1-Score, 88.02% Recall, 6.7k params).
3. **Compiled Report:** Refer to [docs/stockout_architectures_and_results.pdf] (stockout_architectures_and_results.pdf) for the updated publication-ready PDF documentation.